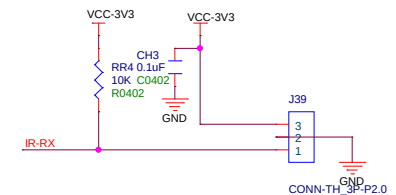


The schematic diagram illustrates the HDMI input section of the PCB. It features two HDMI connectors, JH1 and JH2, each with 19 pins. The signals from these connectors are routed through two identical HDMI input modules, ULC0524P RCLAMP0524P, which are connected to a 5V supply and ground. The output signals from these modules are connected to the HDMI input pins of the HDMI controller (HDMI19+). The diagram also shows the connection of the HDMI input pins to the HDMI controller pins, including the hot plug detect (HPD) signal.

IR-RX



Design Name			
H133_DONGLE_STD			
Size	Page Name	Rev	
A3	13 USB CARD HDMI		
Date:	Thursday, Jul 30, 2026	Sheet	1 of 14

VERSION HISTORY

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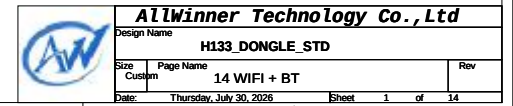
- P01 VERSION HISTORY
- P02 BLOCK DIAGRAM
- P03 POWER TREE
- P04 GPIO ASSIGNMENT
- P05 POWER
- P06 SOC1
- P07 SOC2
- P08 DDR3
- P09 FLASH
- P10 AUDIO
- P11 USB CARD HDMI
- P12 WIFI BT
- OPTIONAL
- P13 DDR2
- P14 USB WIFI

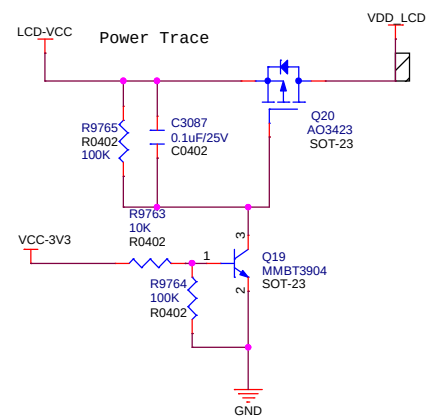
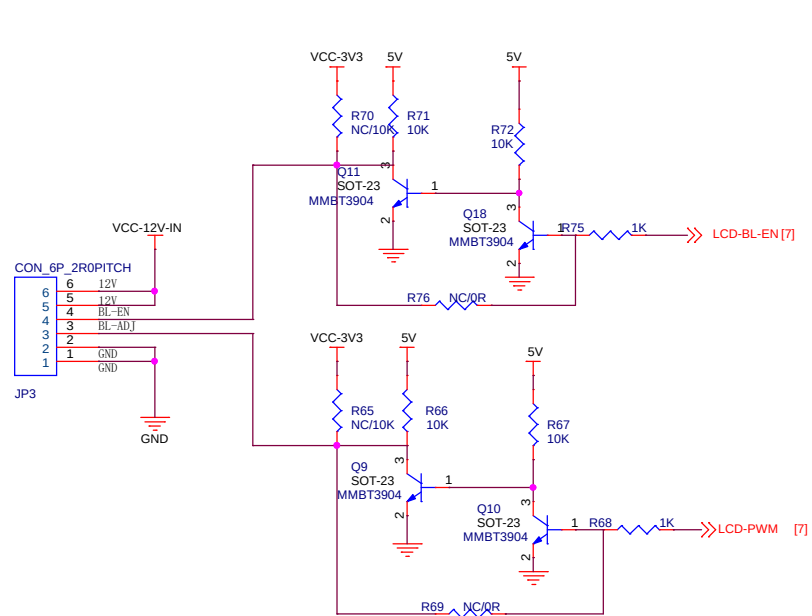
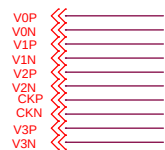
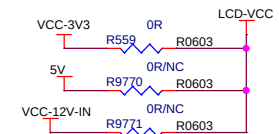
Revision	Description	Date	Drawn	Checked	Approved
Ver 0.1	Releas version	2020-12-31			
Ver 1.0	AddAW859 Add UART2 /5 and SPI test point	2021-04-07	YJY	JIAYONG	YINWEI
Ver 1.1	ADD DDR2 ADD FALE FEL KEY ADD USB WIFI ADD DCDC4 for dram	2021-11-21	WJH	YJY	YINWEI
Ver 1.2	Merge VDD-CPU and vdd-SYS VCC-DRAM change to ext DCDC	2022-01-05	WJH	YJY	YINWEI

[illegible]

The schematic diagram illustrates the electrical connections for the BL-M8731BU4 module. Key components and connections include:

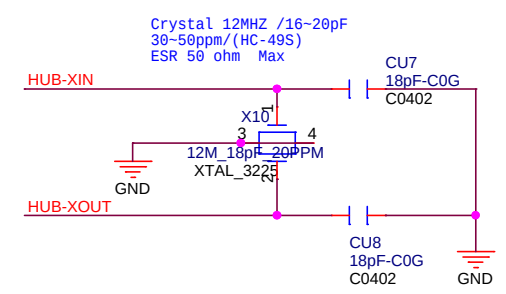
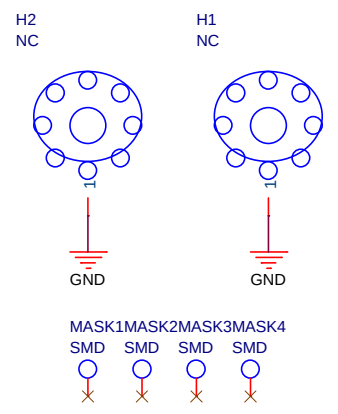
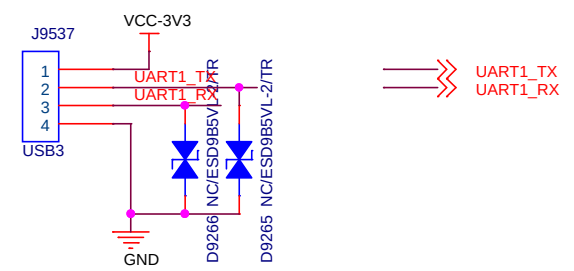
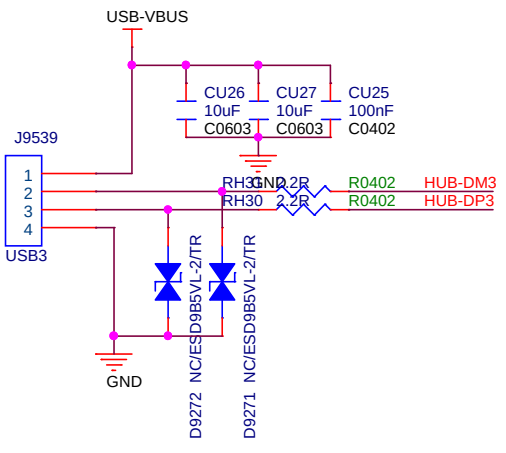
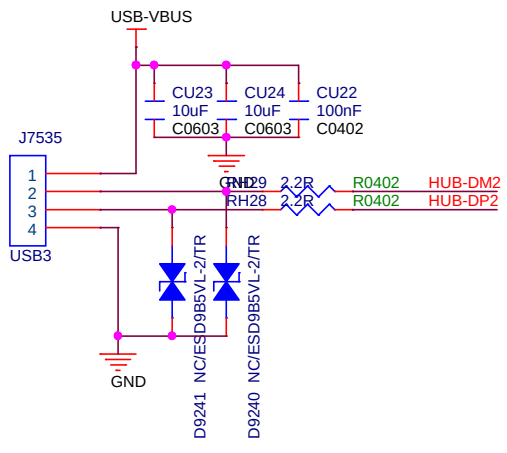
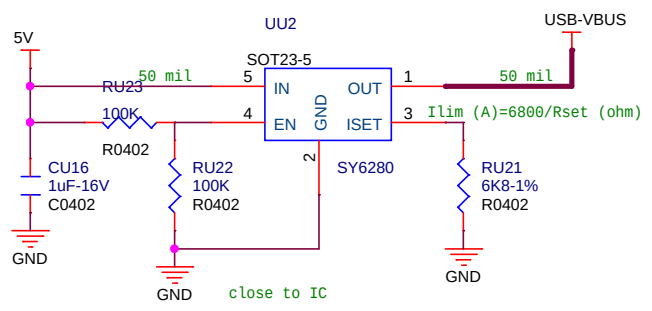
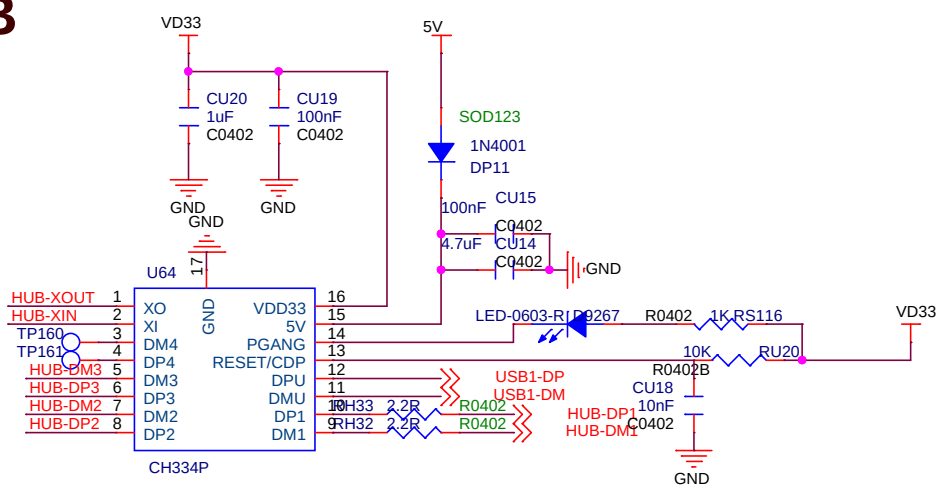
- ANT3 ANT-3PIN:** Connected to the module's antenna pins (1, 2, 3) via a 3-pin connector.
- Reserved for Dual Ant:** A section of the circuit board is marked as reserved for a dual antenna configuration.
- Capacitors:** C16 (NC C0402) and C17 (NC C0402) are connected to ground. C18 (22uF T20603) and C6 (0.1uF C0402) are connected to the VCC33-WIFI pin and ground.
- Grounding:** Multiple ground connections are shown, including GND, GND, and GND.
- Module Pins:** The module is labeled BL-M8731BU4 and includes pins for WL_WAKE_HOST, RF_S1, RF_S1, AGND, BT_PCM_IN, BT_PCM_OUT, BT_PCM_SYNC, BT_WAKE_HOST, HOST_WAKE_BT, TXD, RXD, AGND, USB-DP, USB-DN, and VDD.
- Debug:** A green line labeled "For debug" connects the module's TXD pin to a ground connection.
- HUB-DP1/HUB-DM1:** A connection point for the HUB-DP1/HUB-DM1 is shown on the right side of the diagram.





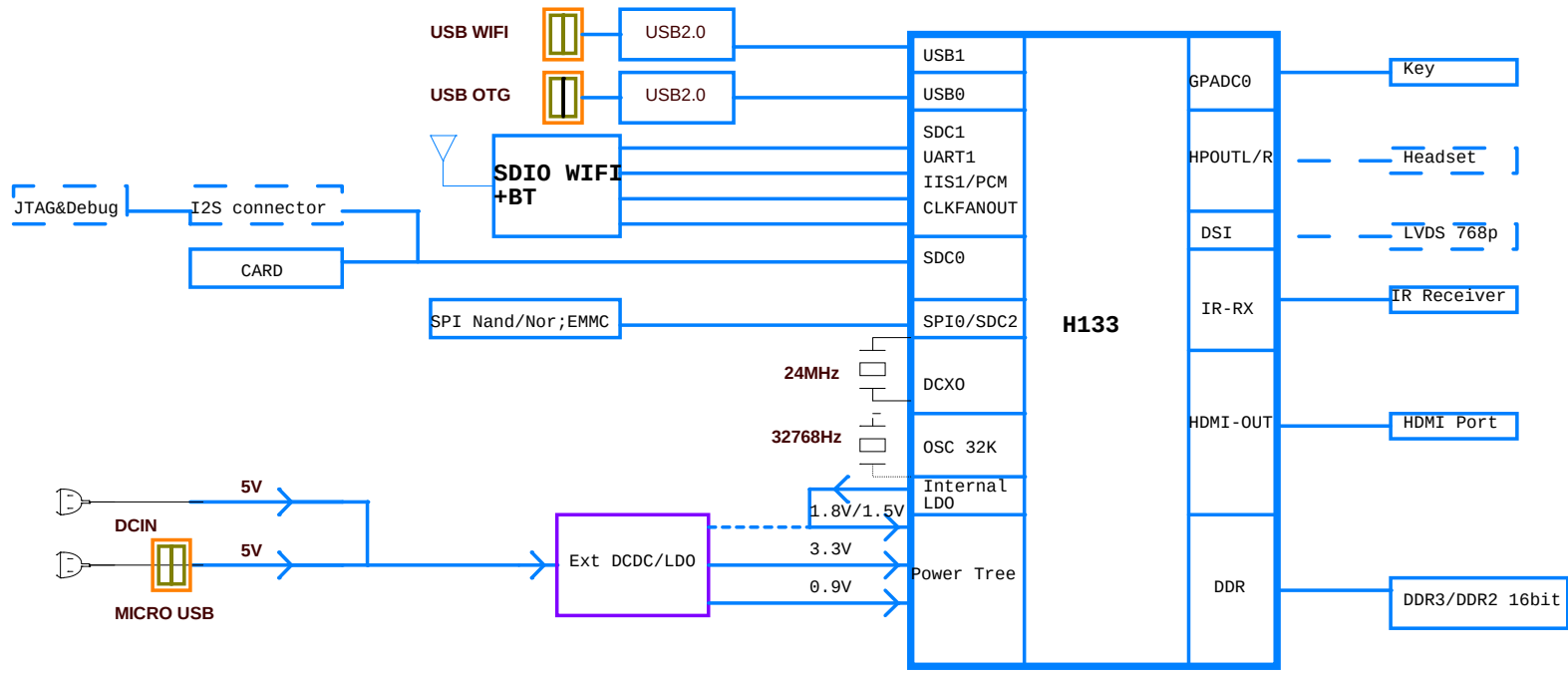
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Size B	Document Number <Doc>	Rev <Rev Code>	
Date:	Thursday, July 30, 2026	Sheet	1 of 1

USB HUB



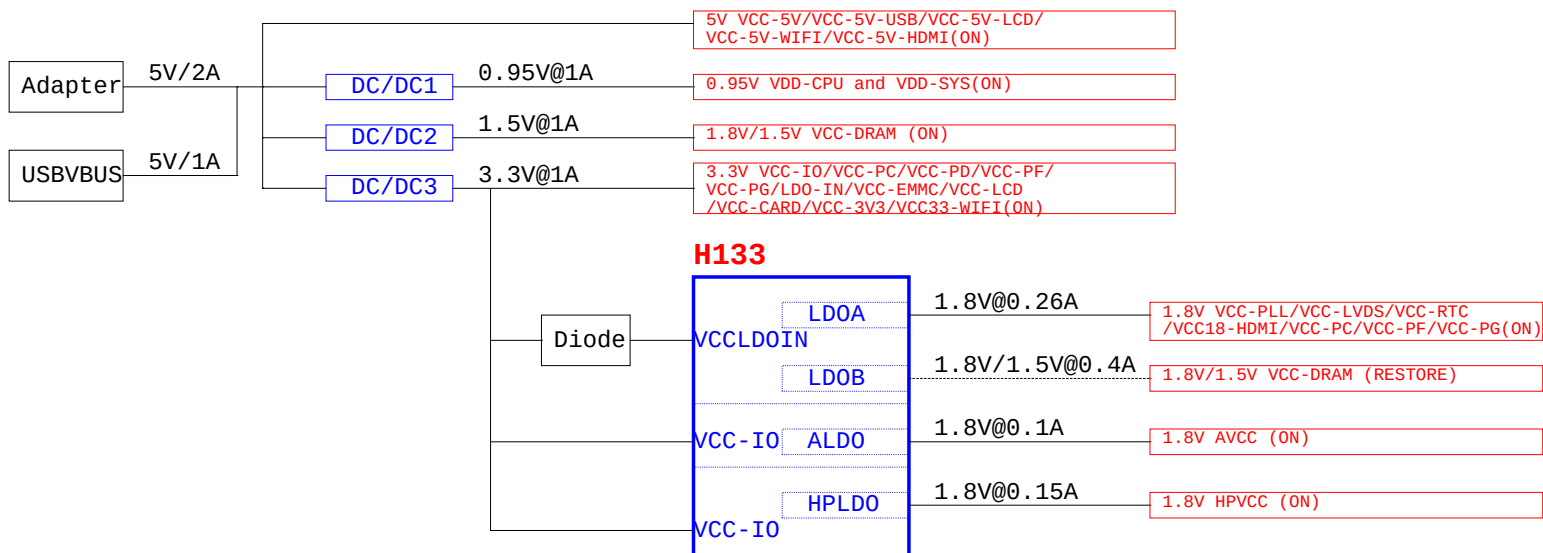
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Size A4		File: USB Port		<RevCode>	
Date: Thursday, July 30, 2026		Date: Thursday, July 30, 2026		Rev: V1.3	
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BLOCK



☐ DEFAULT POWER ON

☐ DEFAULT POWER OFF



GPIO ASSIGNMENT

Ball Number	Ball Name	GPIO Multiplex Function
D15	PB0	PWM3/IR_TX/TWI2_SCK/SPI1_WP/DBI_TE/UART0_TX/UART2_TX/SPDIF_OUT/PB_EINT0
D14	PB1	PWM4/I2S2_DOUT3/TWI2_SDA/I2S2_DIN3/UART0_RX/UART2_RX/IR_RX/PB_EINT1
D13	PB8	DMIC_DATA3/PWM5/TWI2_SCK/SPI1_HOLD/DBI_DCX/DBI_WRX/UART0_TX/UART1_TX/PB_EINT8
C14	PB9	DMIC_DATA2/PWM6/TWI2_SDA/SPI1_MISO/DBI_SDI/DBI_TE/DBI_DCX/UART0_RX/UART1_RX/PB_EINT9
C13	PB10	DMIC_DATA1/PWM7/TWI0_SCK/SPI1_MOSI/DBI_SDO/CLK_FANOUT0/UART1_RTS/PB_EINT10
B15	PB11	DMIC_DATA0/PWM2/TWI0_SDA/SPI1_CLK/DBI_SCLK/CLK_FANOUT1/UART1_CTS/PB_EINT11
B14	PB12	DMIC_CLK/PWM0/SPDIF_IN/SPI1_CS/DBI_CSX/CLK_FANOUT2/IR_RX/PB_EINT12

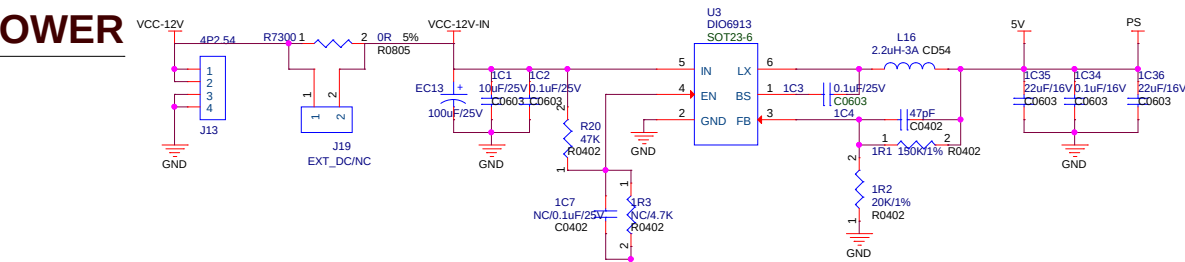
Ball Number	Ball Name	GPIO Multiplex Function
F3	PC2	SPI0_CLK/SDC2_CLK/PC_EINT2
F2	PC3	SPI0_CS0/SDC2_CMD/PC_EINT3
F1	PC4	SPI0_MOSI/SDC2_D2/B00T_SEL0/PC_EINT4
G3	PC5	SPI0_MISO/SDC2_D1/B00T_SEL1/PC_EINT5
G2	PC6	SPI0_WP/SDC2_D0/UART3_TX/TWI3_SCK/DBG_CLK/PC_EINT6
H3	PC7	SPI0_HOLD/SDC2_D3/UART3_RX/TWI3_SDA/TCON_TRIG/PC_EINT7

Ball Number	Ball Name	GPIO Multiplex Function
N15	PD0	LCD0_D2/LVDS0_V0P/DSI_D0P/TWI0_SCK/PD_EINT0
N14	PD1	LCD0_D3/LVDS0_V0N/DSI_D0N/UART2_TX/PD_EINT1
M15	PD2	LCD0_D4/LVDS0_V1P/DSI_D1P/UART2_RX/PD_EINT2
M14	PD3	LCD0_D5/LVDS0_V1N/DSI_D1N/UART2_RTS/PD_EINT3
L15	PD4	LCD0_D6/LVDS0_V2P/DSI_CKP/UART2_CTS/PD_EINT4
L14	PD5	LCD0_D7/LVDS0_V2N/DSI_CKN/UART5_TX/PD_EINT5
K15	PD6	LCD0_D10/LVDS0_CKP/DSI_D2P/UART5_RX/PD_EINT6
K14	PD7	LCD0_D11/LVDS0_CKN/DSI_D2N/UART4_TX/PD_EINT7
J15	PD8	LCD0_D12/LVDS0_V3P/DSI_D3P/UART4_RX/PD_EINT8
J14	PD9	LCD0_D13/LVDS0_V3N/DSI_D3N/PWM6/PD_EINT9

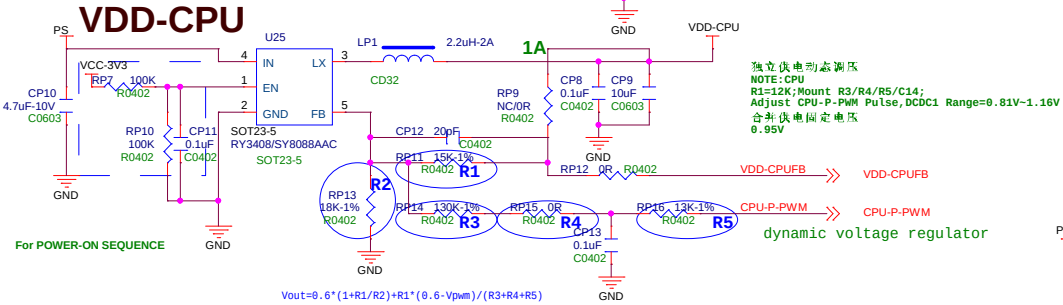
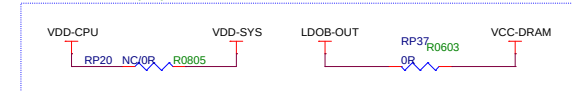
Ball Number	Ball Name	GPIO Multiplex Function
B8	PG0	SDC1_CLK/UART3_TX/RGMII_RXCTRL/RMII_CRS_DV/PWM7/PG_EINT0
C9	PG1	SDC1_CMD/UART3_RX/RGMII_RXD0/RMII_RXD0/PWM6/PG_EINT1
A8	PG2	SDC1_D0/UART3_RTS/RGMII_RXD1/RMII_RXD1/UART4_TX/PG_EINT2
B7	PG3	SDC1_D1/UART3_CTS/RGMII_TXCK/RMII_TXCK/UART4_RX/PG_EINT3
A6	PG4	SDC1_D2/UART5_TX/RGMII_TXD0/RMII_TXD0/PWM5/PG_EINT4
C7	PG5	SDC1_D3/UART5_RX/RGMII_TXD1/RMII_TXD1/PWM4/PG_EINT5
B4	PG6	UART1_TX/TWI2_SCK/RGMII_TXD2/PWM1/PG_EINT6
A3	PG7	UART1_RX/TWI2_SDA/RGMII_TXD3/SPDIF_IN/PG_EINT7
B3	PG8	UART1_RTS/TWI1_SCK/RGMII_RXD2/UART3_TX/PG_EINT8
A2	PG9	UART1_CTS/TWI1_SDA/RGMII_RXD3/UART3_RX/PG_EINT9
C4	PG10	PWM3/TWI3_SCK/RGMII_RXCK/CLK_FANOUT0/IR_RX/PG_EINT10
B6	PG11	I2S1_MCLK/TWI3_SDA/EPHY_25M/CLK_FANOUT1/TCON_TRIG/PG_EINT11
C6	PG12	I2S1_LRCK/TWI0_SCK/RGMII_TXCTRL/RMII_TXEN/CLK_FANOUT2/PWM0/UART1_TX/PG_EINT12
B5	PG13	I2S1_BCLK/TWI0_SDA/RGMII_CLKIN/RMII_RXER/PWM2/LEDC_D0/UART1_RX/PG_EINT13
C5	PG14	I2S1_DIN0/TWI2_SCK/MDC/I2S1_DOUT1/SPI0_WP/UART1_RTS/PG_EINT14
A4	PG15	I2S1_DOUT0/TWI2_SDA/MDIO/I2S1_DIN1/SPI0_HOLD/UART1_CTS/PG_EINT15
B2	PG16	IR_RX/TCON_TRIG/PWM5/CLK_FANOUT2/SPDIF_IN/LEDC_D0/PG_EINT16
C10	PG17	UART2_TX/TWI3_SCK/PWM7/CLK_FANOUT0/IR_TX/UART0_TX/PG_EINT17
B9	PG18	UART2_RX/TWI3_SDA/PWM6/CLK_FANOUT1/SPDIF_OUT/UART0_RX/PG_EINT18

Ball Number	Ball Name	GPIO Multiplex Function
B1	PF0	SDC0_D1/JTAG_MS/R_JTAG_MS/I2S2_DOUT1/I2S2_DIN0/PF_EINT0
C3	PF1	SDC0_D0/JTAG_DI/R_JTAG_DI/I2S2_DOUT0/I2S2_DIN1/PF_EINT1
C2	PF2	SDC0_CLK/UART0_TX/TWI0_SCK/LEDC_D0/SPDIF_IN/PF_EINT2
D3	PF3	SDC0_CMD/JTAG_D0/R_JTAG_D0/I2S2_BCLK/PF_EINT3
D2	PF4	SDC0_D3/UART0_RX/TWI0_SDA/PWM6/IR_TX/PF_EINT4
D1	PF5	SDC0_D2/JTAG_CK/R_JTAG_CK/I2S2_LRCK/PF_EINT5
E2	PF6	SPDIF_OUT/IR_RX/I2S2_MCLK/PWM5/PF_EINT6

POWER



If merge VDD-CPU and VDD-SYS
The both voltage must be fixed at 0.95V
And the CPU-Frequency will NOT above 900Mhz

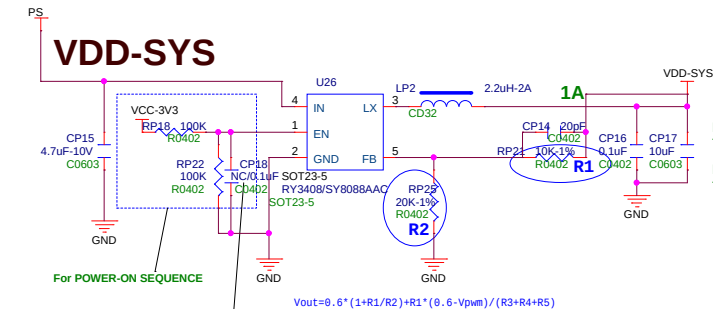


For POWER-ON SEQUENCE

$$V_{out} = 0.6 * (1 + R1/R2) + R1 * (0.6 - V_{pwm}) / (R3 + R4 + R5)$$

独立供电动态调压
NOTE: CPU
R1=12K; Mount R3/R4/R5/C14;
Adjust CPU-P-PWM Pulse, DCDC1 Range=0.81V~1.16V
合并供电固定电压 0.95V

dynamic voltage regulator

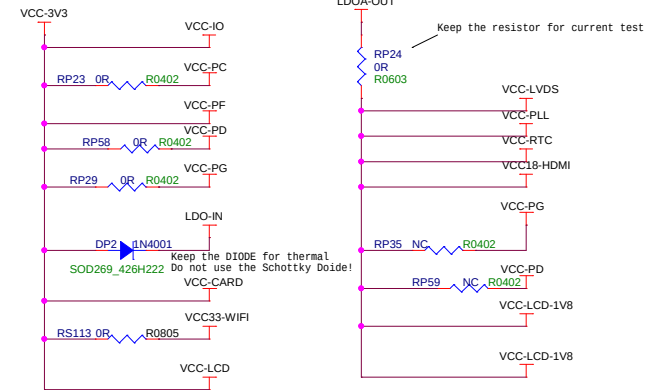


For POWER-ON SEQUENCE

$$V_{out} = 0.6 * (1 + R1/R2) + R1 * (0.6 - V_{pwm}) / (R3 + R4 + R5)$$

此DCDC给VCC-DRAM供电, CP18要NC, 才能满足时序要求

POWER-LED

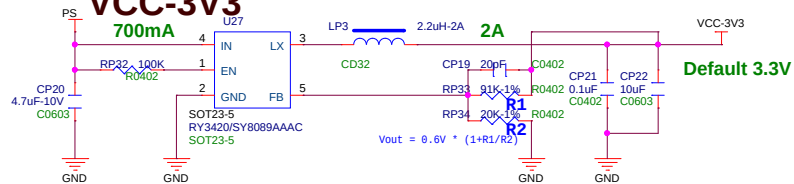


Keep the resistor for current test

Keep the DIODE for thermal
Do not use the Schottky Diode!

	DCDC1 (EXT)	DCDC2 (EXT)	DCDC3 (EXT)	DCDC4 (EXT)	LDOB (IN)	LDOA (IN)	REMARK
Default PLAN	VDD-CPU VDD-SYS (0.95V)	VCC-DRAM 1.5/1.8V	VCC-WIFI VCC-3V3 LDOIN	NC	NC	VCC-PLL VCC-RTC VCC-1V8	Merge VDD-CPU and VDD-SYS ,that Cpu-frequency below 900Mhz.The Power consume is below2.5W@4K30,And the thermal will be very low.
Performance PLAN	VDD-CPU (0.8- 1.16V)	VDD-SYS 0.9V	VCC-WIFI VCC-3V3 LDOIN	VCC-DRAM 1.5/1.8V	NC	VCC-PLL VCC-RTC VCC-1V8	Separate VDD-CPU and VDD-SYS ,The Cpu-frequency upon to 1.2Ghz .The Power consume is below2.5W@4K30,And the thermal will be very low.
Economy PLAN	VDD-CPU VDD-SYS (0.95V)	NC	VCC-WIFI VCC-3V3 LDOIN	NC	VCC-DRAM 1.5/1.8V	VCC-PLL VCC-RTC VCC-1V8	Merge VDD-CPU and VDD-SYS ,that Cpu-frequency below 900Mhz.VCC-Dram be supplied by internal LDOB,instead of Ext DCDC.

VCC-3V3



Default 3.3V

$$V_{out} = 0.6 * (1 + R1/R2)$$

SOC1

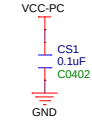
U11C

PC

PC2/SPI0_CLK/SDC2_CLK/PC_EINT2
PC3/SPI0_CS0/SDC2_CMD/PC_EINT3
PC4/SPI0_MOSI/SDC2_D2/BOOT_SEL0/PC_EINT4
PC5/SPI0_MISO/SDC2_D1/BOOT_SEL1/PC_EINT5
PC6/SPI0_WP/SDC2_D0/UART3_TX/TWI3_SCK/DBG_CLK/PC_EINT6
PC7/SPI0_HOLD/SDC2_D3/UART3_RX/TWI3_SDA/TCON_TRIG/PC_EINT7
VCC_PC

3.3V / 1.8V

F3 PC2 RS1 33R R0402 eMMC-CLK SPI0-CLK
F2 F1 eMMC-CMD SPI0-CS0 SPI0-CLK
F1 F1 eMMC-D2 SPI0-MOSI SPI0-CS0
G3 G2 eMMC-D1 SPI0-MISO SPI0-MOSI
H3 H3 eMMC-D0 SPI0-WP SPI0-MISO
F5 F5 eMMC-D3 SPI0-HOLD SPI0-WP
VCC-PC

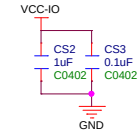


PB

PB0/PWM3/IR_TX/TWI2_SCK/SPI1_WP/DBI_TE/UART0_TX/UART2_TX/SPDIF_OUT/PB_EINT0
PB1/PWM4/I2S2_DOUT3/TWI2_SDA/I2S2_DIN3/UART0_RX/UART2_RX/IR_RX/PB_EINT1
PB8/DMIC_DATA3/PWM5/TWI2_SCK/SPI1_HOLD/DBI_DCX/DBI_WRX/UART0_TX/UART1_TX/PB_EINT8
PB9/DMIC_DATA2/PWM6/TWI2_SDA/SPI1_MISO/DBI_SDI/DBI_TE/DBI_DCX/UART0_RX/UART1_RX/PB_EINT9
PB10/DMIC_DATA1/PWM7/TWI0_SCK/SPI1_MOSI/DBI_SDO/CLK_FANOUT0/UART1_RTS/PB_EINT10
PB11/DMIC_DATA0/PWM2/TWI0_SDA/SPI1_CLK/DBI_SCLK/CLK_FANOUT1/UART1_CTS/PB_EINT11
PB12/DMIC_CLK/PWM0/SPDIF_IN/SPI1_CS/DBI_CSX/CLK_FANOUT2/IR_RX/PB_EINT12
VCC_IO

3.3V

D15 CPU-P-PWM
D14 IR-RX
D13 UART0-TX
C14 UART0-TX
C13 TP157
B15 LCD-RST
B14 LCD-PWM
F11 VCC-IO

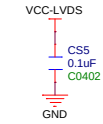
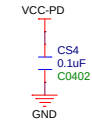


PD

PD0/LCD0_D2/LVDS0_V0P/DSI_D0P/TWI0_SCK/PD_EINT0
PD1/LCD0_D3/LVDS0_V0N/DSI_D0N/UART2_TX/PD_EINT1
PD2/LCD0_D4/LVDS0_V1P/DSI_D1P/UART2_RX/PD_EINT2
PD3/LCD0_D5/LVDS0_V1N/DSI_D1N/UART2_RTS/PD_EINT3
PD4/LCD0_D6/LVDS0_V2P/DSI_D2P/UART2_CTS/PD_EINT4
PD5/LCD0_D7/LVDS0_V2N/DSI_D2N/UART5_TX/PD_EINT5
PD6/LCD0_D10/LVDS0_CK/PD_SCK/D2P/UART5_RX/PD_EINT6
PD7/LCD0_D11/LVDS0_CKN/DSI_D2N/UART4_TX/PD_EINT7
PD8/LCD0_D12/LVDS0_V3P/DSI_D3P/UART4_RX/PD_EINT8
PD9/LCD0_D13/LVDS0_V3N/DSI_D3N/PWM6/PD_EINT9
VCC_LVDS
VCC_PD

3.3V

N15 LVDS-V0P DSI-D0P LVDS-V0P
N14 LVDS-V0N DSI-D0N LVDS-V0N
M15 LVDS-V1P DSI-D1P LVDS-V1P
M14 LVDS-V1N DSI-D1N LVDS-V1N
L15 LVDS-V2P DSI-D2P LVDS-V2P
L14 LVDS-V2N DSI-D2N LVDS-V2N
K15 LVDS-CKP DSI-D2P LVDS-CKP
K14 LVDS-CKN DSI-D2N LVDS-CKN
J15 LVDS-V3P DSI-D3P LVDS-V3P
J14 LVDS-V3N DSI-D3N LVDS-V3N
G11 VCC-LVDS
J11 VCC-PD



PF

PF0/SDC0_D1/JTAG_MS/R_JTAG_MS/I2S2_DOUT1/I2S2_DIN0/PF_EINT0
PF1/SDC0_D0/JTAG_DI/R_JTAG_DI/I2S2_DOUT0/I2S2_DIN1/PF_EINT1
PF2/SDC0_CLK/UART0_TX/TWI0_SCK/LEDC_DO/SPDIF_IN/PF_EINT2
PF3/SDC0_CMD/JTAG_DO/R_JTAG_DO/I2S2_BCLK/PF_EINT3
PF4/SDC0_D3/UART0_RX/TWI0_SDA/PWM6/IR_TX/PF_EINT4
PF5/SDC0_D2/JTAG_CK/R_JTAG_CK/I2S2_LRCK/PF_EINT5
PF6/SPDIF_OUT/IR_RX/I2S2_MCLK/PWM5/PF_EINT6
VCC_PF

3.3V

B1 PF0 SDC0-D1 SDC0-D1
C3 PF1 SDC0-D0 SDC0-D0
C2 RS115 33R R0402 SDC0-D0
D3 PF3 SDC0-CMD SDC0-CLK
D2 PF4 SDC0-D3 SDC0-CMD
D1 PF5 SDC0-D2 SDC0-D3
E2 PF6 SDC0-DET SDC0-D2
E5 VCC-PF SDC0-DET



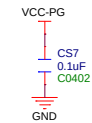
U11G

PG

PG0/SDC1_CLK/UART3_TX/RGMII_RXCTRL/RMII_CRS_DV/PWM7/PG_EINT0
PG1/SDC1_CMD/UART3_RX/RGMII_RXD0/RMII_RXD0/PWM6/PG_EINT1
PG2/SDC1_D0/UART3_RTS/RGMII_RXD1/RMII_RXD1/UART4_TX/PG_EINT2
PG3/SDC1_D1/UART3_CTS/RGMII_TXCK/RMII_TXCK/UART4_RX/PG_EINT3
PG4/SDC1_D2/UART5_TX/RGMII_TXD0/RMII_TXD0/PWM5/PG_EINT4
PG5/SDC1_D3/UART5_RX/RGMII_TXD1/RMII_TXD1/PWM4/PG_EINT5
PG6/UART1_TX/TWI2_SCK/RGMII_TXD2/PWM1/PG_EINT6
PG7/UART1_RX/TWI2_SDA/RGMII_TXD3/SPDIF_IN/PG_EINT7
PG8/UART1_RTS/TWI2_SCK/RGMII_RXD2/UART3_TX/PG_EINT8
PG9/UART1_CTS/TWI1_SDA/RGMII_RXD3/UART3_RX/PG_EINT9
PG10/PWM3/TWI3_SCK/RGMII_RXCK/CLK_FANOUT0/IR_RX/PG_EINT10
PG11/I2S1_MCLK/TWI3_SDA/EPHY_25MCLK_FANOUT1/TCON_TRIG/PG_EINT11
PG12/I2S1_LRCK/TWI0_SCK/RGMII_TXCTRL/RMII_TXENCLK_FANOUT2/PWM0/UART1_TX/PG_EINT12
PG13/I2S1_BCLK/TWI0_SDA/RGMII_CLKIN/RMII_RXER/PWM2/LEDC_DO/UART1_RX/PG_EINT13
PG14/I2S1_DIN0/TWI2_SCK/MDC/I2S1_DOUT1/SPI0_WP/UART1_RTS/PG_EINT14
PG15/I2S1_DOUT0/TWI2_SDA/MDC/I2S1_DIN1/SPI0_HOLD/UART1_CTS/PG_EINT15
PG16/IR_RX/TCON_TRIG/PWM5/CLK_FANOUT2/SPDIF_IN/LEDC_DO/PG_EINT16
PG17/UART2_TX/TWI3_SCK/PWM7/CLK_FANOUT0/IR_TX/UART0_TX/PG_EINT17
PG18/UART2_RX/TWI3_SDA/PWM6/CLK_FANOUT1/SPDIF_OUT/UART0_RX/PG_EINT18
VCC_PG

1.8V / 3.3V

B8 UART_TX
C9 UART_RX
A8 UART-RX
B7 CTP-RST
A6 LED-PWM [13]
C7 TP158
B4 UART1_TX
A3 UART1_RX
B3 UART1_RX
A2 CTP-SCK
C4 CTP-SDA
B6 I2S1-MCLK 14
C6 I2S1-LRCLK 14
B5 I2S1-BCLK 14
C5 I2S1-DIN0 14
A4 I2S1-DOUT0 14
B2 DC_DET [13]
C10 CD-DET [13]
B9 CD-BL-EN
D5 SPDIF-OUT [18]
VCC-PG
CF1 100PF
C0402
GND



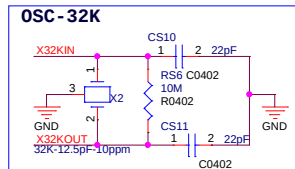
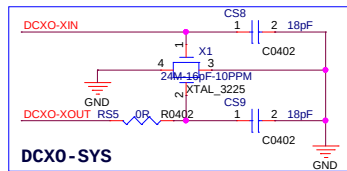
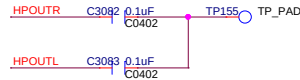
H133-BGA196

BGA196P6581000_1000H117

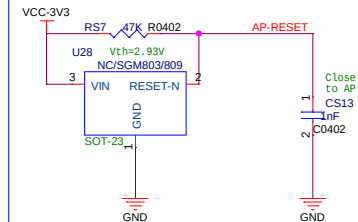
Allwinner Technology Co.,Ltd			
Design Name		H133_DONGLE_STD	
Size	Page Name	Rev	
A3	07 SOC1		
Date:	Thursday, July 30, 2026	Sheet	6 of 12

SOC2

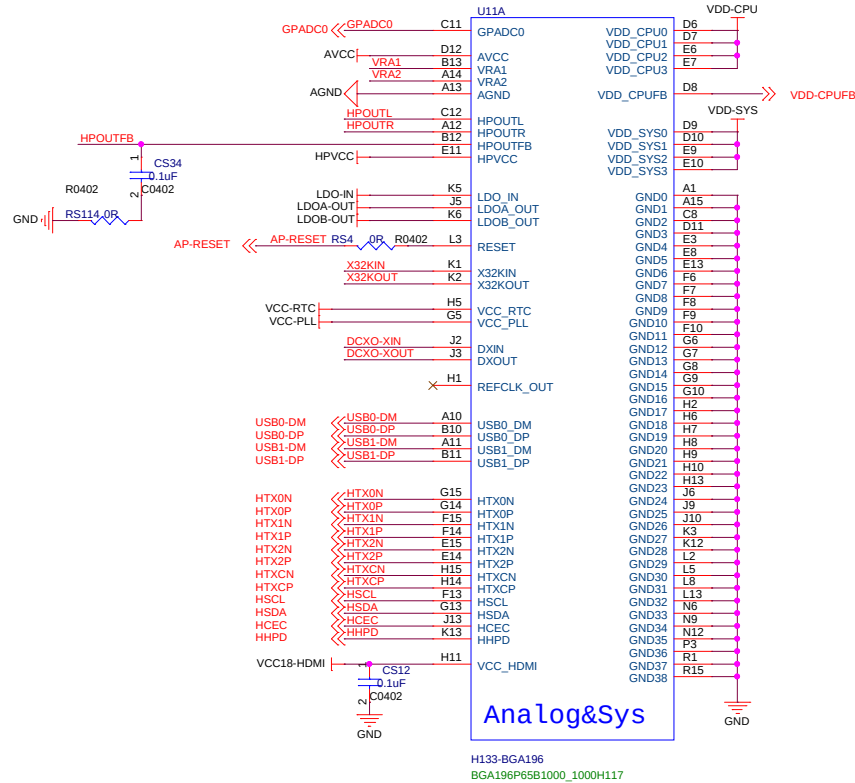
Audio



RESET NOTE: RESET IC can NC, When Using SoC Internal RESET Logic.
信号远离板边以及干扰信号, 做包地处理

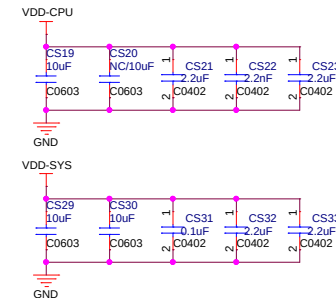
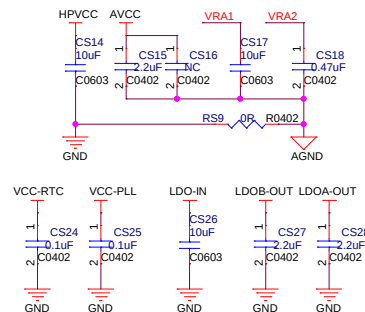


When use the SPI FLASH.
Keep the SGM803 to reduce the Flash Data lose,
when Abnormal Poweroff



Analog&Sys

H133-BGA196
BGA196P65B1000_1000H117



DDR3

Pinout for H133-BGA196:

Pin	Signal	Pin	Signal
SDQ0	L4	SA0	N11
SDQ1	M4	SA1	M11
SDQ2	M5	SA2	M10
SDQ3	M6	SA3	L10
SDQ4	N4	SA4	L11
SDQ5	N5	SA5	R13
SDQ6	P5	SA6	M9
SDQ7	P4	SA7	P13
SDQ8	R3	SA8	P12
SDQ9	M1	SA9	P11
SDQ10	P2	SA10	N10
SDQ11	M2	SA11	P10
SDQ12	N3	SA12	R14
SDQ13	N2	SA13	L12
SDQ14	R2	SA14	P15
SDQ15	M3	SA15	M12
SDQS0P	P6	SDQS0N	SBA0
SDQS1P	R6	SDQS1N	SBA1
SDQM0	L6	SDQM1	R4
SVREF	K11	SVREF	K10

NOTE: 1. VDD18-DRAM bonding to VCC-PLL

Pinout for FBGA78P0_8B7_5X11:

Pin	Signal	Pin	Signal
SA0	K3	SDQ0	B3
SA1	L7	SDQ1	C7
SA2	L3	SDQ2	C2
SA3	K2	SDQ3	A2
SA4	L8	SDQ4	A3
SA5	L2	SDQ5	A4
SA6	M8	SDQ6	A5
SA7	M2	SDQ7	A6
SA8	N8	SDQ8	A7
SA9	M3	SDQ9	A8
SA10	H7	SDQ10	A9
SA11	M7	SDQ11	A10
SA12	K7	SDQ12	A11
SA13	N3	SDQ13	A12
SA14	N7	SDQ14	A13
SA15	J7	SDQ15	A14
SDQS0P	C3	SDQS0N	D3
SDQM0	B7	SDQM1	D7
SVREF	E1	SVREF	J8

Pinout for FBGA78P0_8B7_5X11:

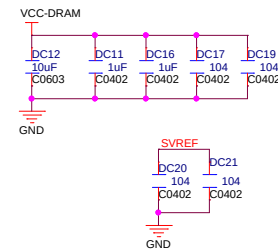
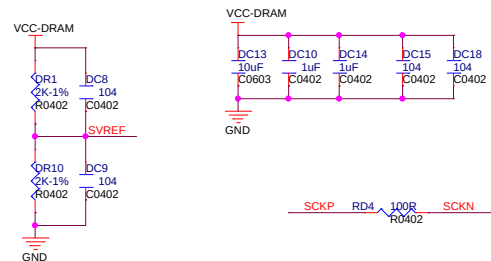
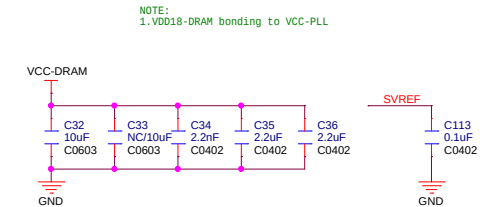
Pin	Signal	Pin	Signal
SA0	K3	SDQ0	B3
SA1	L7	SDQ1	C7
SA2	L3	SDQ2	C2
SA3	K2	SDQ3	A2
SA4	L8	SDQ4	A3
SA5	L2	SDQ5	A4
SA6	M8	SDQ6	A5
SA7	M2	SDQ7	A6
SA8	N8	SDQ8	A7
SA9	M3	SDQ9	A8
SA10	H7	SDQ10	A9
SA11	M7	SDQ11	A10
SA12	K7	SDQ12	A11
SA13	N3	SDQ13	A12
SA14	N7	SDQ14	A13
SA15	J7	SDQ15	A14
SDQS0P	C3	SDQS0N	D3
SDQM0	B7	SDQM1	D7
SVREF	E1	SVREF	J8

Pinout for FBGA78P0_8B7_5X11:

Pin	Signal	Pin	Signal
SA0	K3	SDQ0	B3
SA1	L7	SDQ1	C7
SA2	L3	SDQ2	C2
SA3	K2	SDQ3	A2
SA4	L8	SDQ4	A3
SA5	L2	SDQ5	A4
SA6	M8	SDQ6	A5
SA7	M2	SDQ7	A6
SA8	N8	SDQ8	A7
SA9	M3	SDQ9	A8
SA10	H7	SDQ10	A9
SA11	M7	SDQ11	A10
SA12	K7	SDQ12	A11
SA13	N3	SDQ13	A12
SA14	N7	SDQ14	A13
SA15	J7	SDQ15	A14
SDQS0P	C3	SDQS0N	D3
SDQM0	B7	SDQM1	D7
SVREF	E1	SVREF	J8

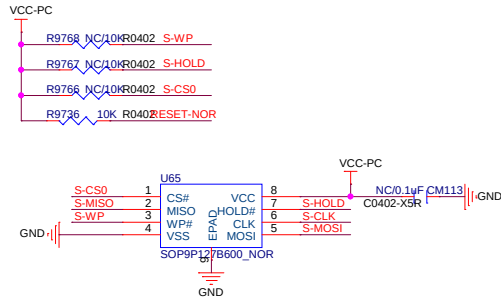
Pinout for FBGA78P0_8B7_5X11:

Pin	Signal	Pin	Signal
SA0	K3	SDQ0	B3
SA1	L7	SDQ1	C7
SA2	L3	SDQ2	C2
SA3	K2	SDQ3	A2
SA4	L8	SDQ4	A3
SA5	L2	SDQ5	A4
SA6	M8	SDQ6	A5
SA7	M2	SDQ7	A6
SA8	N8	SDQ8	A7
SA9	M3	SDQ9	A8
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SA11	M7	SDQ11	A10
SA12	K7	SDQ12	A11
SA13	N3	SDQ13	A12

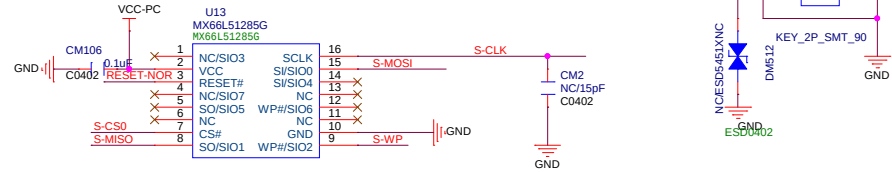


SPI

SPI0-CLK << S-CLK
 SPI0-CS0 << S-CS0
 SPI0-MOSI << S-MOSI
 SPI0-MISO << S-MISO
 SPI0-WP << S-WP
 SPI0-HOLD << S-HOLD



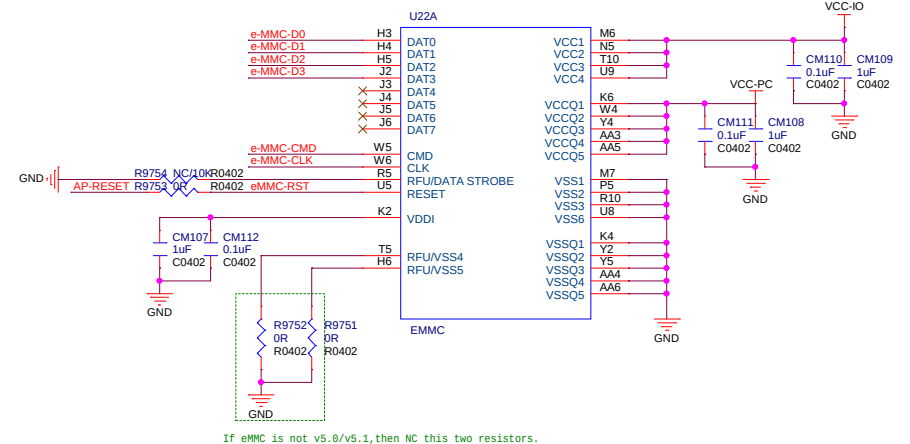
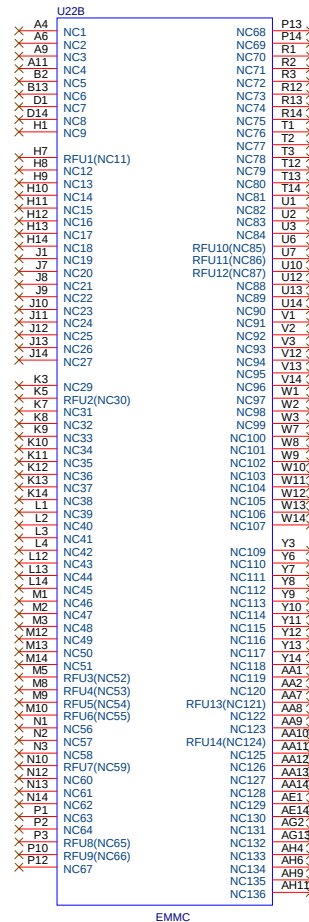
UBOOT



EMMC

SPI FLASH and EMMC only can choose one of them.
 Both Layout in PCB will effect the SI

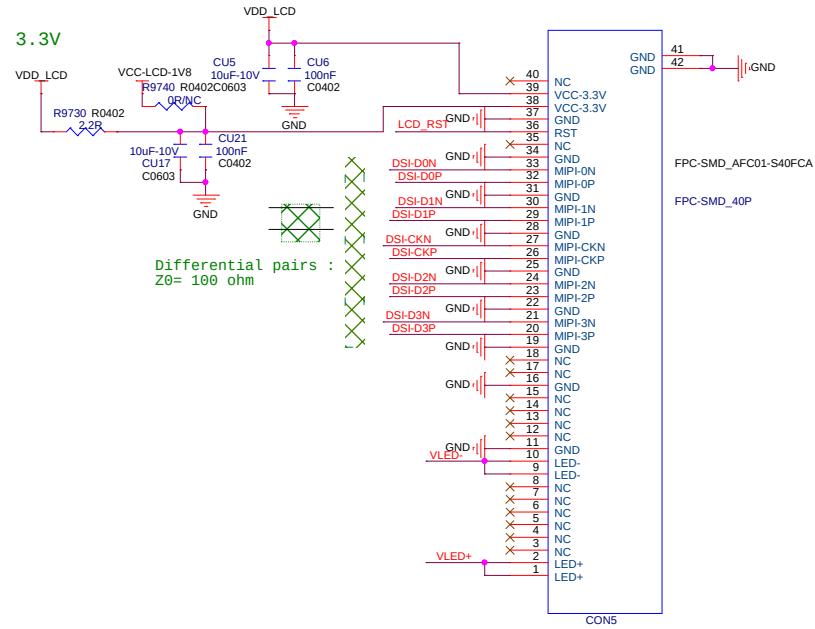
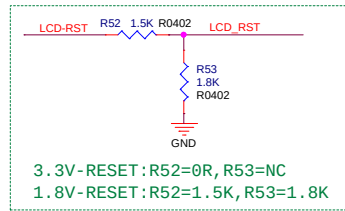
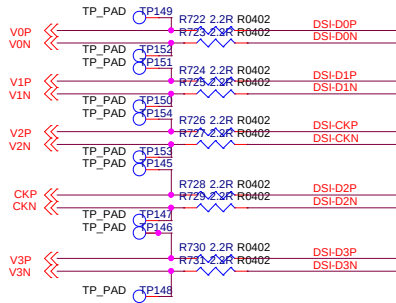
SPI0-CLK << R9748 OR/NB0402 eMMC-CLK
 SPI0-CS0 << R9746 OR/NB0402 eMMC-CMD
 SPI0-MOSI << R9747 OR/NB0402 eMMC-D2
 SPI0-MISO << R9748 OR/NB0402 eMMC-D1
 SPI0-WP << R9749 OR/NB0402 eMMC-D0
 SPI0-HOLD << R9750 OR/NB0402 eMMC-D3
 AP-RESET << AP-RESET



If eMMC is not v5.0/v5.1, then NC this two resistors.

LCD

3.3V

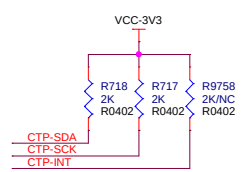
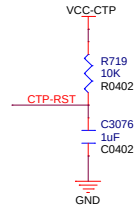
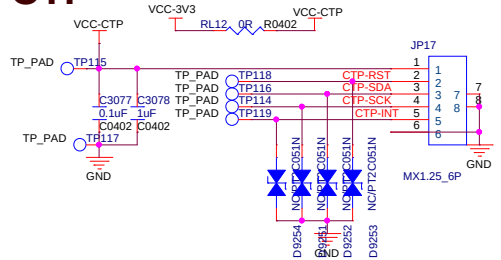


Differential pairs :
 Z0= 100 ohm

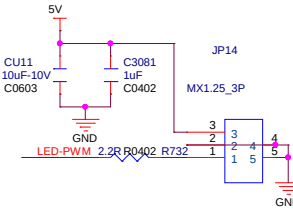
VCC-LCD
 I_{MAX}=0.2A

PS
 I_{MAX}=0.5A

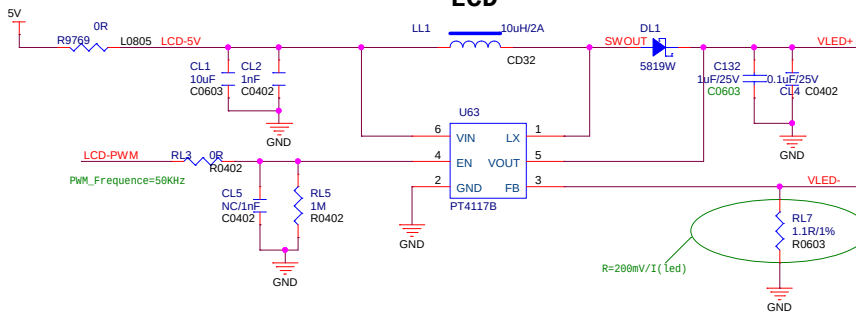
CTP



氛围灯



Power Trace



BACKLIGHT



Allwinner Technology Co., Ltd			
Design Name			
A100_A133_B3_AXP717_LPDDR4			
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